Letters to the Editor

Diamond Burr Polishing

Dear Editor:
The authors report that outcomes after diamond burr polishing are better than those after phototherapeutic keratectomy (PTK) when treating recurrent corneal erosions (RCEs).[1] This is good news to all of the ophthalmologists and patients who do not have access to an excimer laser.

The unfavorable results reported after excimer laser treatments, though disappointing, are nonetheless intriguing. As the authors mention in their discussion, PTK has been generally effective in the treatment of RCEs. Our own experience with PTK has also been positive,[2] even when treating recurrent erosions in bullous keratopathy.[3]

I have comments on two problems mentioned in this article:

1. We have encountered mild corneal haze in only one patient after treatment. In that case no subjective reduction of visual function was observed by the patient. It should be mentioned that some patients with RCE do have some haze before treatment. Slit-lamp examination of the cornea can reveal where corneal erosions will start. The authors do not mention if they looked for corneal haze preoperatively. The fact that this study is retrospective and not masked weakens the authors' conclusions.
2. It is common and effective practice in the treatment of RCEs to debride most of the epithelium using a sponge alone. This is effective because the adhesion between the epithelium and the underlying Bowman's layer is very loose, not only where a loose epithelium is being observed, but also many times on the whole corneal surface. Because most commercial excimer lasers have a maximal ablation diameter of 10 mm, it is extremely important to treat the periphery of the erosion. (The VISX, which had been used by these authors, has a maximal ablation diameter of 9 mm.) The authors should mention the diameter they used, and whether they extended the area treated to include the periphery.

We use computer-controlled ablation in the central 10 mm and treat the periphery manually. Our technique is not only in a nonoverlapping manner, as described by the authors, but in an overlapping manner as well, including an overlapping of the epithelium. This assures equivalent good adhesion in the periphery. We have had no recurrences since using this technique.

ISAAC SCHIPPER, MD
Lucerne, Switzerland

References

1. Sridhar MS, Rapuano CJ, Cosar CB, et al. Phototherapeutic keratectomy versus diamond burr polishing of Bowman's membrane in the treatment of recurrent corneal erosions associated with anterior basement membrane dystrophy. Ophthalmology 2002;109:674–9.
2. Niesen U, Thomann U, Schipper I. Phototherapeutic keratectomy [in German]. Klin Monatsbl Augenheilkd 1994;205:187–95.
3. Thomann U, Meier-Gibbons F, Schipper I. Phototherapeutic keratectomy for bullous keratopathy. Br J Ophthalmol 1995; 79:335–8.

Author reply

Dear Editor:
Dr. Schipper writes that we concluded "that outcomes after diamond burr polishing are better than those after phototherapeutic keratectomy (PTK)." We did not state this. To paraphrase our conclusions, we stated that both treatments are effective; however, diamond burr polishing, being simpler and less expensive, may have advantages over PTK.

Dr. Schipper notes correctly that patients may have preexisting corneal haze before a diamond burr or PTK procedure. We did not systematically look for preoperative corneal haze, as many patients had their procedure during an episode of recurrent erosion where haze would be difficult to quantify. As noted in the article, we did not find a significant difference in postoperative haze between the two groups.

Dr. Schipper discussed his technique of PTK, where he uses a 10-mm ablation. He states that the VISX laser (VISX Inc, Santa Clara, CA) we used has a 9-mm maximal ablation capability. We clearly discuss the exact PTK technique in the second paragraph of Materials and Methods. The VISX laser models we used (20/20 B and Star S2, as stated in that section) have maximal PTK ablation diameters of 6.0 mm. We described our technique of debriding the loose epithelium and treating with the laser. If the epithelial defect was 6 mm in diameter or smaller, we used a single large-zone ablation of 6 mm. If the epithelial defect was larger than 6 mm in diameter, then the central 6-mm area was treated and then masked with a 6-mm-diameter filter paper and the periphery was treated with nonoverlapping 3- to 4-mm-diameter ablations. Consequently, the entire epithelial defect would be treated just once for 5 μm. The outer treatments did overlap the peripheral intact epithelium.

We continue to use both diamond burr polishing and PTK as part of our armamentarium in the management of recurrent erosion syndrome.

CHRISTOPHER J. RAPUANO, MD
Philadelphia, Pennsylvania

Corneal Refractive Power after Myopic LASIK

Dear Editor:
The recent article by Hamed et al[1] addresses an emerging problem that many ophthalmologists will be facing in the relatively near future. It has two main conclusions: (1) when manual keratometry and topography are used for intraocular lens (IOL) calculations after myopic LASIK, the power is underestimated, and (2) there is a direct, linear relationship between the degree of corneal refractive surgery and IOL power error as evidenced by regression analysis. This study also makes several important recommendations, the most notable of which is reducing the keratometry readings by 0.24 diopters (D) per diopter of refractive correction and